

Continuity, Smoothness, and Differentiability

David Puelz

The University of Austin

Outline

1. **Continuity** – Definition and intuition (15 min)
2. **Types of Discontinuities** – Classification and examples (15 min)
3. **Differentiability** – When derivatives exist (15 min)
4. **Smoothness Classes** – The C^n hierarchy (15 min)
5. **Theorems Connecting Concepts** – Key results, proofs, and why they matter (15 min)

1 Continuity

1.1 Intuitive Definition

The pencil test: A function is **continuous** if you can draw its graph without lifting your pencil from the paper.

More precisely: A function is continuous at a point if a small change in the input produces only a small change in the output—no sudden jumps.

1.2 Formal Definition

Definition 1.1. A function f is **continuous at** $x = a$ if:

$$\lim_{x \rightarrow a} f(x) = f(a)$$

This requires three things:

1. $f(a)$ exists (the function is defined at a)
2. $\lim_{x \rightarrow a} f(x)$ exists (the limit exists)
3. These two values are equal

Definition 1.2. A function is **continuous on an interval** if it is continuous at every point in that interval.

1.3 The ϵ - δ Definition

Definition 1.3 (ϵ - δ formulation). A function f is continuous at a if for every $\epsilon > 0$, there exists a $\delta > 0$ such that:

$$|x - a| < \delta \quad \Rightarrow \quad |f(x) - f(a)| < \epsilon$$

Interpretation:

- ϵ is how close we want outputs to be
- δ is how close inputs need to be to guarantee this
- Continuity means we can always find such a δ

1.4 Examples of Continuous Functions

All pattern-book functions (the nine from Lecture 1) are continuous on their domains:

- Exponential e^x , sinusoid $\sin(x)$, $\cos(x)$: continuous everywhere
- Logarithm $\ln(x)$: continuous for $x > 0$
- Reciprocal $1/x$: continuous for $x \neq 0$
- Square x^2 , identity x , one: continuous everywhere
- Gaussian e^{-x^2} , sigmoid $\frac{1}{1+e^{-x}}$: continuous everywhere

Combinations preserve continuity:

- Sum/difference of continuous functions is continuous
- Product of continuous functions is continuous
- Quotient is continuous where denominator $\neq 0$
- Composition of continuous functions is continuous

2 Types of Discontinuities

2.1 Removable Discontinuity

Definition 2.1. A **removable discontinuity** occurs when the limit exists but either:

- The function is not defined at that point, or
- The function value differs from the limit

Example:

$$f(x) = \frac{x^2 - 1}{x - 1} = \frac{(x - 1)(x + 1)}{x - 1}$$

For $x \neq 1$: $f(x) = x + 1$

At $x = 1$: undefined (but $\lim_{x \rightarrow 1} f(x) = 2$)

The discontinuity is “removable” because we could define $f(1) = 2$ to make f continuous.

2.2 Jump Discontinuity

Definition 2.2. A **jump discontinuity** occurs when the left and right limits both exist but are not equal:

$$\lim_{x \rightarrow a^-} f(x) \neq \lim_{x \rightarrow a^+} f(x)$$

Classic example: The Heaviside step function

$$H(x) = \begin{cases} 0 & \text{if } x < 0 \\ 1 & \text{if } x \geq 0 \end{cases}$$

At $x = 0$:

- $\lim_{x \rightarrow 0^-} H(x) = 0$
- $\lim_{x \rightarrow 0^+} H(x) = 1$
- Jump size: $1 - 0 = 1$

Applications:

- Switching circuits (on/off)
- Tax brackets (different rates at thresholds)
- Phase transitions (water freezing)

2.3 Infinite Discontinuity

Definition 2.3. An **infinite discontinuity** (or **vertical asymptote**) occurs when the function approaches $\pm\infty$ as x approaches some value.

Example: $f(x) = \frac{1}{x}$ at $x = 0$

- $\lim_{x \rightarrow 0^+} \frac{1}{x} = +\infty$
- $\lim_{x \rightarrow 0^-} \frac{1}{x} = -\infty$

Example: $f(x) = \tan(x)$ at $x = \pi/2$

2.4 Oscillating Discontinuity

Definition 2.4. An **oscillating discontinuity** occurs when the function oscillates infinitely rapidly near a point, so no limit exists.

Example:

$$f(x) = \sin\left(\frac{1}{x}\right)$$

As $x \rightarrow 0$, the argument $1/x \rightarrow \pm\infty$, causing $\sin(1/x)$ to oscillate between -1 and $+1$ infinitely often. No limit exists.

2.5 Summary of Discontinuity Types

Type	Characteristic	Example
Removable	Limit exists, function undefined or wrong	$\frac{x^2-1}{x-1}$ at $x = 1$
Jump	Left and right limits differ	Heaviside at $x = 0$
Infinite	Function $\rightarrow \pm\infty$	$\frac{1}{x}$ at $x = 0$
Oscillating	No limit exists	$\sin(1/x)$ at $x = 0$

3 Differentiability

3.1 Definition

Definition 3.1. A function f is **differentiable at** $x = a$ if the derivative $f'(a)$ exists, i.e., if:

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

exists and is finite.

Definition 3.2. A function is **differentiable on an interval** if it is differentiable at every point in that interval.

3.2 Differentiability Implies Continuity

The proof uses one fact about limits.

Lemma 3.3 (Limit of a product). If $\lim_{x \rightarrow a} A(x) = L$ and $\lim_{x \rightarrow a} B(x) = M$, then $\lim_{x \rightarrow a} (A(x) \cdot B(x)) = L \cdot M$.

Theorem 3.4. If f is differentiable at $x = a$, then f is continuous at $x = a$.

Proof (step-by-step):

Step 1. We must show that f is continuous at a . By definition, that means we must show $\lim_{x \rightarrow a} f(x) = f(a)$.

Step 2. Equivalently, we show that $\lim_{x \rightarrow a} (f(x) - f(a)) = 0$. (If this limit is zero, then $\lim_{x \rightarrow a} f(x) = f(a)$.)

Step 3. For $x \neq a$, we can write

$$f(x) - f(a) = \frac{f(x) - f(a)}{x - a} \cdot (x - a).$$

This is valid because we are not dividing by zero when $x \neq a$.

Step 4. Take the limit as $x \rightarrow a$ on both sides. The right-hand side is a product of two expressions:

- $A(x) = \frac{f(x) - f(a)}{x - a}$. By definition of the derivative, $\lim_{x \rightarrow a} A(x) = f'(a)$.
- $B(x) = x - a$. We have $\lim_{x \rightarrow a} B(x) = \lim_{x \rightarrow a} (x - a) = 0$.

Step 5. By Lemma 3.3 (limit of a product),

$$\lim_{x \rightarrow a} (f(x) - f(a)) = \lim_{x \rightarrow a} \left(\frac{f(x) - f(a)}{x - a} \cdot (x - a) \right) = f'(a) \cdot 0 = 0.$$

Step 6. So $\lim_{x \rightarrow a} (f(x) - f(a)) = 0$, which gives $\lim_{x \rightarrow a} f(x) = f(a)$.

Therefore f is continuous at a . $\square$

Important: The converse is **false**! Continuity does NOT imply differentiability.

3.3 Continuous but Not Differentiable: The Absolute Value

Consider $f(x) = |x|$:

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

Continuity at $x = 0$:

- $\lim_{x \rightarrow 0^+} |x| = 0$
- $\lim_{x \rightarrow 0^-} |x| = 0$
- $|0| = 0$

All equal, so f is continuous at 0. $\checkmark$

Differentiability at $x = 0$:

$$\text{Right derivative: } \lim_{h \rightarrow 0^+} \frac{|0 + h| - |0|}{h} = \lim_{h \rightarrow 0^+} \frac{h}{h} = 1$$

$$\text{Left derivative: } \lim_{h \rightarrow 0^-} \frac{|0 + h| - |0|}{h} = \lim_{h \rightarrow 0^-} \frac{-h}{h} = -1$$

Since $1 \neq -1$, the limit doesn't exist. The function has a **corner** at $x = 0$.

Conclusion: $|x|$ is continuous everywhere but not differentiable at $x = 0$.

3.4 Other Ways Differentiability Can Fail

1. *Corners:* Where the graph has a sharp point (like $|x|$).
2. *Cusps:* Where the tangent becomes vertical from both sides.
Example: $f(x) = x^{2/3}$ at $x = 0$

$$f'(x) = \frac{2}{3}x^{-1/3} = \frac{2}{3\sqrt[3]{x}}$$

As $x \rightarrow 0$, $f'(x) \rightarrow \pm\infty$. The curve has a vertical tangent (cusp).

3. *Vertical tangents:* Where the derivative is infinite.
Example: $f(x) = \sqrt[3]{x}$ at $x = 0$

$$f'(x) = \frac{1}{3}x^{-2/3} \rightarrow +\infty \text{ as } x \rightarrow 0$$

4. *Discontinuities:* Any discontinuity prevents differentiability (by the theorem).

4 Smoothness Classes

4.1 The C^n Classification

Definition 4.1. A function belongs to class C^n on an interval if:

- C^0 : f is continuous
- C^1 : f' exists and is continuous
- C^2 : f'' exists and is continuous
- C^n : $f^{(n)}$ exists and is continuous
- C^∞ : All derivatives exist and are continuous

Hierarchy:

$$C^\infty \subset \dots \subset C^2 \subset C^1 \subset C^0$$

A function in C^n is automatically in C^{n-1} , C^{n-2} , etc.

4.2 Examples

C^∞ functions (smooth):

- All polynomials
- e^x , $\sin(x)$, $\cos(x)$
- $\ln(x)$ for $x > 0$
- Sums, products, compositions of C^∞ functions

C^0 but not C^1 :

- $f(x) = |x|$ (corner at origin)
- Piecewise linear functions with corners

C^1 but not C^2 : Consider:

$$f(x) = \begin{cases} x^2 & \text{if } x \geq 0 \\ -x^2 & \text{if } x < 0 \end{cases}$$

Then:

$$f'(x) = \begin{cases} 2x & \text{if } x \geq 0 \\ -2x & \text{if } x < 0 \end{cases} = 2|x|$$

f' is continuous (so $f \in C^1$), but $f'' = 2\operatorname{sgn}(x)$ has a jump at $x = 0$ (so $f \notin C^2$).

4.3 Why Smoothness Matters

1. *Physics*: Position should be C^2 (acceleration defined). Ideally C^3 (jerk defined) for smooth motion.
2. *Computer graphics*: Curves connecting points should be C^2 to avoid visible kinks.
3. *Numerical methods*: Higher smoothness allows better approximations (Taylor series).
4. *Differential equations*: Solution existence often requires certain smoothness.

5 Theorems Connecting the Concepts

5.1 Extreme Value Theorem

Theorem 5.1 (Extreme Value Theorem). *If f is continuous on a closed interval $[a, b]$, then f attains both a maximum and minimum value on $[a, b]$.*

Intuitive proof: Picture the graph of f on $[a, b]$. It's a connected curve with no gaps (continuity) and no escape to infinity (closed interval). The curve is “trapped” in a finite horizontal strip. There must be a highest point and a lowest point—and because the curve has no jumps, it actually *touches* those heights. If the curve only approached a maximum without ever reaching it, we could walk along the interval finding points that get higher and higher; eventually we'd have to “land” somewhere (the interval is closed), and continuity ensures the value there is the maximum we were approaching. The minimum works the same way.

Why both conditions matter:

- **Closed interval:** On $(0, 1]$, the function $f(x) = 1/x$ is continuous but has no maximum— $f(x) \rightarrow \infty$ as $x \rightarrow 0^+$.
- **Continuity:** The function $f(x) = x$ on $[0, 1)$ has no maximum—it approaches 1 but never reaches it.

Why it matters: Whenever we ask “what’s the best (or worst) that can happen?”—profit maximization, cost minimization, worst-case analysis—we’re looking for extrema. The EVT guarantees that on a closed interval, such extrema *exist*, so our optimization problem has a solution. Without it, we might search forever.

5.2 Intermediate Value Theorem

Theorem 5.2 (Intermediate Value Theorem). *If f is continuous on $[a, b]$ and N is any number between $f(a)$ and $f(b)$, then there exists $c \in (a, b)$ such that $f(c) = N$.*

Intuitive proof: Imagine you’re hiking from elevation $f(a)$ to elevation $f(b)$. Your path is continuous—no teleporting, no jumps. If N is any elevation between $f(a)$ and $f(b)$, you must pass through elevation N at some point. There’s no way to get from one height to another without crossing every height in between. Formally: bisect the interval. If the midpoint’s value equals N , we’re done. Otherwise, N lies between the values at the endpoints of one half; repeat. The nested intervals shrink to a point c , and continuity forces $f(c) = N$.

Why it matters:

- **Finding roots:** If $f(a) < 0$ and $f(b) > 0$, then 0 is between $f(a)$ and $f(b)$, so f has a root in (a, b) . This justifies bisection and many numerical root-finding methods.

- **Existence without formulas:** We know a solution exists even when we can't write it in closed form (e.g., $x^5 - 3x + 1 = 0$).
- **Fixed points:** If g is continuous on $[0, 1]$ and $g([0, 1]) \subseteq [0, 1]$, then g has a fixed point—IVT applied to $f(x) = g(x) - x$.

Example: $f(x) = x^3 - x - 1$. We have $f(1) = -1 < 0$ and $f(2) = 5 > 0$. By IVT, there exists $c \in (1, 2)$ with $f(c) = 0$.

5.3 Rolle's Theorem

Theorem 5.3 (Rolle's Theorem). *If f is continuous on $[a, b]$, differentiable on (a, b) , and $f(a) = f(b)$, then there exists $c \in (a, b)$ with $f'(c) = 0$.*

Proof (Rolle's Theorem): We use a lemma: if f has a local maximum or minimum at $c \in (a, b)$ and $f'(c)$ exists, then $f'(c) = 0$. (At a peak, the difference quotient from the left is ≥ 0 and from the right is ≤ 0 ; so $f'(c) \geq 0$ and $f'(c) \leq 0$, hence $f'(c) = 0$.)

Case 1: f is constant on $[a, b]$. Then $f'(x) = 0$ for all $x \in (a, b)$; any $c \in (a, b)$ works.

Case 2: f is not constant. By the Extreme Value Theorem, f attains a maximum M and minimum m on $[a, b]$. Since $f(a) = f(b)$ and f is not constant, at least one of $M > f(a)$ or $m < f(a)$ holds. So f attains a max or min at some $c \in (a, b)$. By the lemma, $f'(c) = 0$. $\square$

Why it matters: Rolle's theorem is the key lemma for proving the Mean Value Theorem and many other results. It captures the idea that "what goes up must come down"—and at the turnaround, the derivative is zero.

5.4 Mean Value Theorem

Theorem 5.4 (Mean Value Theorem). *If f is continuous on $[a, b]$ and differentiable on (a, b) , then there exists $c \in (a, b)$ such that:*

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

Proof (Mean Value Theorem): Define the auxiliary function

$$g(x) = f(x) - f(a) - \frac{f(b) - f(a)}{b - a}(x - a).$$

Then g is continuous on $[a, b]$ and differentiable on (a, b) . We have

$$g(a) = f(a) - f(a) - 0 = 0 \quad \text{and} \quad g(b) = f(b) - f(a) - (f(b) - f(a)) = 0.$$

So $g(a) = g(b)$. By Rolle's Theorem, there exists $c \in (a, b)$ such that $g'(c) = 0$. Since

$$g'(x) = f'(x) - \frac{f(b) - f(a)}{b - a},$$

we obtain $g'(c) = 0 \Rightarrow f'(c) = \frac{f(b) - f(a)}{b - a}$. $\square$

Interpretation: Somewhere between a and b , the instantaneous rate of change equals the average rate of change. On a road trip, your average speed must equal your instantaneous speed at some moment (usually when you're speeding).

Why it matters:

- **Monotonicity:** If $f' > 0$ on an interval, then f is increasing (MVT shows $f(b) - f(a) = f'(c)(b - a) > 0$).
- **Error bounds:** Taylor's theorem and many numerical methods rely on MVT for remainder estimates.
- **Fundamental Theorem:** MVT is used to prove that antiderivatives differ by constants.

Why both conditions matter:

- Continuity on $[a, b]$ ensures no gaps (otherwise the "average" could be misleading).
- Differentiability on (a, b) ensures the derivative exists where we need it.

6 Summary

6.1 Key Relationships

Differentiable $\Rightarrow$ Continuous $\Rightarrow$ Defined

But the converses are false!

6.2 Smoothness Hierarchy

$$C^\infty \subset \dots \subset C^2 \subset C^1 \subset C^0$$

6.3 Practice Exercises

Exercise 1: Classify the discontinuity (if any) at $x = 0$ for:

(a) $f(x) = \frac{\sin x}{x}$

(b) $g(x) = \frac{1}{x^2}$

(c) $h(x) = \frac{|x|}{x}$

Exercise 2: Determine the smoothness class (C^0 , C^1 , C^2 , or C^∞) of:

$$f(x) = \begin{cases} x^3 & \text{if } x \geq 0 \\ -x^3 & \text{if } x < 0 \end{cases}$$

Exercise 3: Prove that $f(x) = x^{1/3}$ is continuous but not differentiable at $x = 0$.

Exercise 4: Use the Intermediate Value Theorem to show that $x^5 - 3x + 1 = 0$ has a solution in $(-1, 0)$.

Exercise 5: Apply the Mean Value Theorem to $f(x) = x^2$ on $[1, 3]$ to find the value c guaranteed by the theorem.

Exercise 6: Construct a function that is C^1 but not C^2 at $x = 1$.

Worked Solutions

Exercise 1 (Solution): Classify the discontinuity at $x = 0$.

- (a) $f(x) = \frac{\sin x}{x}$: At $x = 0$, f is not defined (o/o). The limit $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$ exists. So this is a **removable discontinuity**—we could define $f(0) = 1$ to make f continuous.
- (b) $g(x) = \frac{1}{x^2}$: As $x \rightarrow 0$, $g(x) \rightarrow +\infty$. So this is an **infinite discontinuity** (vertical asymptote at $x = 0$).
- (c) $h(x) = \frac{|x|}{x}$: For $x > 0$, $h(x) = 1$; for $x < 0$, $h(x) = -1$. So $\lim_{x \rightarrow 0^+} h(x) = 1$ and $\lim_{x \rightarrow 0^-} h(x) = -1$. The left and right limits exist but differ. This is a **jump discontinuity**.

□

Exercise 2 (Solution): $f(x) = x^3$ for $x \geq 0$ and $f(x) = -x^3$ for $x < 0$. Then $f'(x) = 3x^2$ for $x > 0$, $f'(x) = -3x^2$ for $x < 0$, and $f'(0) = 0$ (both one-sided derivatives equal 0). So $f'(x) = 3x|x|$, which is continuous. Hence $f \in C^1$. For $x \neq 0$, $f''(x) = 6x$ for $x > 0$ and $f''(x) = -6x$ for $x < 0$, so $f''(x) = 6|x|$. At $x = 0$, f'' is continuous (limit from both sides is 0). So f'' is continuous everywhere, hence $f \in C^2$. For $x > 0$, $f'''(x) = 6$; for $x < 0$, $f'''(x) = -6$. So f''' has a jump at $x = 0$ and is not continuous there. **Conclusion:** f is C^2 but not C^3 . □

Exercise 3 (Solution): $f(x) = x^{1/3}$.

Continuity at 0: $\lim_{x \rightarrow 0} x^{1/3} = 0 = f(0)$, so f is continuous at 0.

Not differentiable at 0: For $x \neq 0$, $f'(x) = \frac{1}{3}x^{-2/3}$. As $x \rightarrow 0$, $|f'(x)| \rightarrow +\infty$. The difference quotient at 0 is $\frac{(0+h)^{1/3}-0}{h} = h^{-2/3}$, which tends to $\pm\infty$ as $h \rightarrow 0$ (and is not defined for $h < 0$ in the reals when we use the real cube root). So the limit does not exist; f is not differentiable at 0. □

Exercise 4 (Solution): Let $f(x) = x^5 - 3x + 1$. Then f is continuous on $\mathbb{R}$. **Step 1:** Check the interval $(-1, 0)$. We have $f(-1) = (-1)^5 - 3(-1) + 1 = -1 + 3 + 1 = 3 > 0$ and $f(0) = 1 > 0$. So f is positive at both endpoints of $(-1, 0)$ —there is no sign change, so IVT does not by itself guarantee a root in $(-1, 0)$. **Step 2:** To apply IVT we need an interval where f takes opposite signs. On $[0, 1]$: $f(0) = 1 > 0$ and $f(1) = 1 - 3 + 1 = -1 < 0$. So 0 is between $f(0)$ and $f(1)$. By the Intermediate Value Theorem, there exists $c \in (0, 1)$ such that $f(c) = 0$. So $x^5 - 3x + 1 = 0$ has a solution in $(0, 1)$. (For a solution in $(-1, 0)$ we would need $f(-1)$ and $f(0)$ to have opposite signs; here

they are both positive, so the standard IVT argument gives a root in $(0, 1)$ instead.) $\square$

Exercise 5 (Solution): For $f(x) = x^2$ on $[1, 3]$, the Mean Value Theorem says there exists $c \in (1, 3)$ such that $f'(c) = \frac{f(3)-f(1)}{3-1} = \frac{9-1}{2} = 4$. Since $f'(x) = 2x$, we need $2c = 4$, so $c = 2$. So the value guaranteed by the theorem is $c = 2$. $\square$

Exercise 6 (Solution): One example: let

$$f(x) = \begin{cases} (x-1)^2 & \text{if } x \geq 1 \\ -(x-1)^2 & \text{if } x < 1 \end{cases}$$

Then $f'(x) = 2(x-1)$ for $x > 1$ and $f'(x) = -2(x-1)$ for $x < 1$, so $f'(x) = 2|x-1|$, and $f'(1) = 0$. So f' is continuous (and f is C^1). But $f''(x) = 2$ for $x > 1$ and $f''(x) = -2$ for $x < 1$, so f'' has a jump at $x = 1$ (left limit -2 , right limit 2). So f'' is not continuous at 1 , hence $f \notin C^2$ at $x = 1$. So f is C^1 but not C^2 at $x = 1$. $\square$